

RESEARCH ARTICLE

Characterization of a Novel RNA Polymerase Mutant of *Escherichia coli* That Confers Thermotolerance and Chaperone Independence

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ABSTRACT

Bacterial cells activate stress-sensing and stress-mitigating pathways by employing a number of transcription regulators, some of which can bind directly to RNA polymerase to activate stress-specific response pathways. However, mutations in the RNA polymerase genes can accumulate under certain selection conditions and activate stress-mitigating pathways in a manner that is partly independent of pathway-specific regulators. In this study, we characterized a novel mutation in the *rpoB* gene that transforms RNA polymerase into a “stringent” polymerase in the absence of one of the key stringent response (SR) activating factors (p)ppGpp, produced by the *relA* gene product. The mutant RNA polymerase allele, *rpoB58*, elevated thermotolerance and permitted growth without the key molecular chaperones (DnaKJ) and proteases (Lon, ClpP) at temperatures nonpermissive to cells expressing the wild type RNA polymerase genes. Remarkably, *rpoB58* also reversed the cell division defect of Δ *dnaJ* at a nonpermissive temperature but could not overcome its lambda phage-resistant phenotype. The *rpoB58*-mediated rescue of the Δ *dnaKJ* growth defect was partly reversed in the absence of DksA, a protein that acts synergistically with (p)ppGpp to transform RNA polymerase into a stringent state. The data suggest that pre-activated SR confers thermotolerance and chaperone independence in part by lowering protein synthesis.

1 | Introduction

Elevating temperature above that which is optimal for growth of a bacterial species causes protein misfolding and an imbalance in protein homeostasis (Schramm et al. 2020). This triggers a heat stress response, which, in *Escherichia coli*, is governed by the σ^{32} (σ^H) subunit of RNA polymerase (Grossman et al. 1984). Synthesis of two main classes of proteins is activated: chaperones and proteases. Together, these proteins act to either lower protein misfolding/aggregation or facilitate their degradation in an attempt to reestablish protein homeostasis. For correct folding, many *E. coli* proteins rely on the two major molecular

chaperone and chaperonin complexes, comprised of DnaKJGrpE and GroELS, respectively (Balchin et al. 2020). In vivo studies have identified several dozen *E. coli* proteins that are substrates of either the DnaKJGrpE complex (Mogk et al. 1992; Calloni et al. 2012) or the GroELS complex (Kerner et al. 2005; Fujiwara et al. 2010). Even though substrates of both complexes include essential proteins, only GroELS and GrpE are essential under all growth conditions. The conditional (temperature-dependent) essentiality of DnaKJ indicates either the existence of redundant chaperone activities or that the level of protein misfolding under permissive growth temperatures ($\leq 30^\circ\text{C}$) is low enough to be tolerated by the cell without DnaKJ. In contrast, increased

protein misfolding without DnaKJ under nonpermissive growth temperatures (> 30°C) likely affects essential cellular functions and/or causes toxic accumulation of protein aggregates.

The maximum growth temperature for wild type (WT) *E. coli* strains has been reported to be between 44.5°C and 46°C (Hoffman and Frank 1963; van Derlinden et al. 2008; Rudolph et al. 2010). Although it is not entirely clear what limits *E. coli*'s growth at these temperatures, there is an increase in protein misfolding and aggregation, and the structure–function aspect of the membrane is likely destabilized at high growth temperatures. The fatty acyl chains of membrane phospholipids become more saturated with increasing growth temperatures and more unsaturated at lower temperatures to fine-tune membrane fluidity, which is critical for maintaining permeability and functioning of the membrane proteins (Zhang and Rock 2008). It has been reported that membranes become more permeable to protons with increasing temperatures (van de Vossenberg et al. 1995). The authors suggested that increased leakiness of the membrane to protons may become growth-limiting at temperatures much higher than the optimal growth temperature of a given bacterial species (van de Vossenberg et al. 1995).

E. coli strains can adapt to grow at temperatures much higher than their optimal growth temperatures, and different strategies have been used to investigate the mechanism of adaptation. Adaptive laboratory evolution (ALE) experiments have been conducted by growing bacterial cultures at either progressively higher temperatures (Kishimoto et al. 2010; Rudolph et al. 2010; Blaby et al. 2012; Kosaka et al. 2019) or at a fixed temperature (Tenaillon et al. 2012) for multiple generations. Regardless of the method of evolving heat-adapted *E. coli* strains, the mechanistic basis for the heat tolerance phenotypes has remained unclear due to the accumulation of a large number of mutations in different genes in the evolved strains and the pleiotropic nature of some mutations (Kishimoto et al. 2010; Tenaillon et al. 2012; Rodríguez-Verdugo et al. 2014; Luan et al. 2015; Kosaka et al. 2019; McGuire and Nano 2023). Nevertheless, included among the mutations identified in these studies were mutations in the genetic loci affecting either the transcription machinery or the stringent response (SR) stress pathway (Kishimoto et al. 2010; Tenaillon et al. 2012; Kosaka et al. 2019). In an entirely different approach to identify genes required for growth at high temperatures, the single gene knockout library of *E. coli* was screened for growth at 47°C (Murata et al. 2011). Not surprisingly, some of the genes identified in the screen code for chaperones (*dnaKJ*) and a protease (*degP*) that are known to be essential for growth at high temperatures (Paek and Walker 1987; Sell et al. 1990; Strauch et al. 1989). Interestingly, the screen also identified *dksA*, a gene coding for a critical SR regulator (Paul et al. 2004). It is worth noting that *dksA* was originally identified in a genetic screen for plasmid clones that overcame the temperature sensitive growth defect of a *dnaK* deletion mutant (Kang and Craig 1990). Thus, three fundamentally different experimental strategies to investigate the mechanism of heat tolerance identified genes affecting SR. A previous study showed the need for SR-proficiency for growth on medium without salt at 42°C (Yang and Ishiguro 2003). These authors also showed that certain “stringent” RNA polymerase mutants (Zhou and Jin 1998) can suppress the temperature sensitive growth phenotype in a stringent-defective background deleted for *relA*, a

gene responsible for the synthesis of the SR alarmone (p)ppGpp (Stent and Brenner 1961; Cashel and Gallant 1969). However, the mechanistic basis of temperature sensitivity without the active SR and its suppression by the “stringent” RNA polymerase mutants could not be established.

In this study, we examined the thermotolerance phenotype of an RNA polymerase mutant in which a mutation in the *rpoB* gene (*rpoB58*) transforms the polymerase into a “stringent-like” state that adopts a gene expression profile resembling the SR (Cho and Misra 2021); we refer to this as the stringent-like response (S-LR). We show that *rpoB58* extends the upper temperature growth limit from 45°C to 47°C in rich medium and renders the growth of the *E. coli* strains independent of the DnaKJ chaperones and Lon and ClpP proteases at temperatures normally lethal for strains carrying the wild type *rpoB* allele. Moreover, our data implicate the protein synthesis-inhibiting branch of SR in thermotolerance and chaperone independence.

2 | Results

2.1 | Thermal Tolerance Conferred by *rpoB58*

Due to their pleiotropic nature, certain mutations in the RNA polymerase genes have been known to positively or negatively influence growth under conditions in which they were not originally selected (Conrad et al. 2010; Dragosits et al. 2013; Utrilla et al. 2016). With this in mind, we tested whether the RNA polymerase mutation *rpoB58*, which was isolated among antibiotic-resistant revertants of a stringent-defective *relA1* strain lacking two major drug efflux pumps (Cho and Misra 2021), can also confer tolerance to unrelated stresses. The *rpoB58* mutation causes a G449V substitution in the β -lobe region of the β -subunit (RpoB) of RNA polymerase. The mutant *rpoB58* allele was moved into an efflux-proficient strain background carrying the *relA1* allele for subsequent examination.

In this work, we tested whether *rpoB58* can confer thermotolerance through its pleiotropic effects. This was initially determined by examining the growth of wild type (here referred to as *rpoBWT*) and *rpoB58* strains on solid medium incubated at different temperatures (Figure 1). Lack of single colonies or the presence of different size (heterogeneous) colonies would indicate the inability of bacteria to tolerate that temperature. At 30°C and 37°C, both strains formed homogeneous colonies, although at 30°C, colonies from *rpoB58* were visibly smaller than *rpoBWT*. At 45°C, this behavior reversed, with *rpoBWT* forming smaller, albeit still homogeneous, colonies than *rpoB58*. *rpoBWT* formed heterogeneous colonies at 46°C, but *rpoB58* maintained homogeneous growth. At 47°C, *rpoBWT* altogether failed to form single colonies, while *rpoB58* formed homogeneous colonies that were slightly smaller than at 46°C. When the incubation temperature was raised to 48°C, *rpoB58* could no longer form homogeneous colonies. These results showed that *rpoB58* increases the thermotolerance of the bacterial strain on solid medium.

We then proceeded to confirm the thermotolerance phenotype of *rpoB58* by conducting liquid growth experiments. Measurements of optical density (OD) were used to reflect

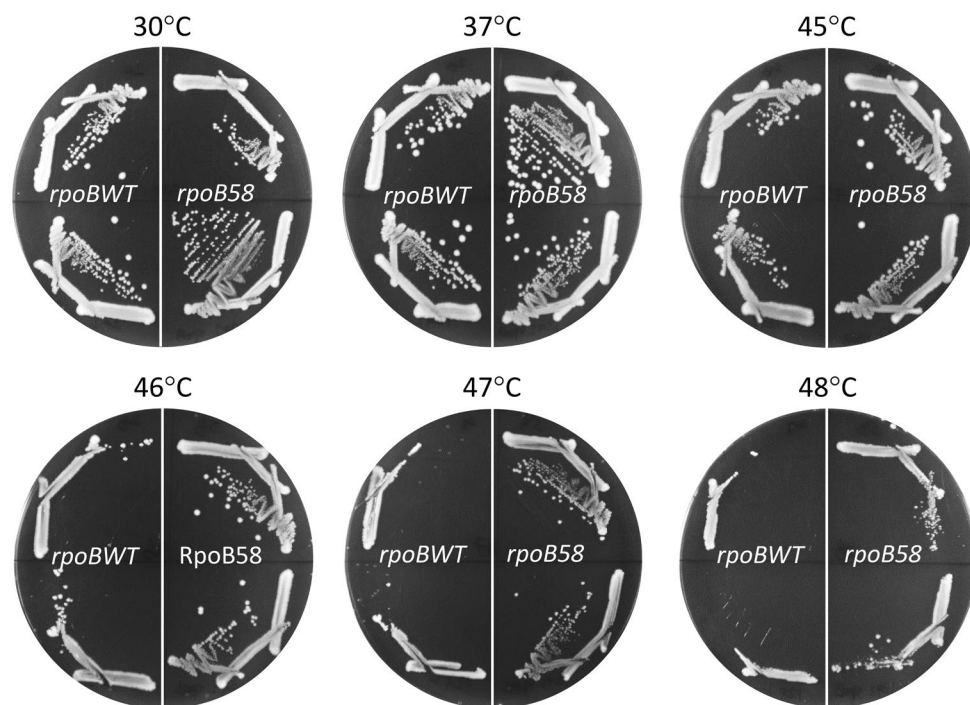


FIGURE 1 | Effect of temperature on single colony formation abilities of *rpoBWT* and *rpoB58* strains. Bacterial growth was recorded on lysogenic broth agar medium after incubating petri plates at temperatures shown for 24 h. Two independent, 37°C-grown colonies of *rpoBWT* and *rpoB58* were tested.

biomass, and measurements of colony forming units (CFUs) were used to reflect viable cell counts. Both *rpoBWT* and *rpoB58* cultures followed a typical growth curve at 37°C, except that during the majority of the growth period, the mutant culture had significantly lower OD readings than *rpoBWT* (Figure 2A). Interestingly, however, samples plated from the two cultures of disparate OD produced almost identical CFUs (Figure 2B). High temperature growth experiments were conducted at 46°C, because at this temperature the mutant displays robust growth on solid medium while *rpoBWT* begins to struggle to form homogeneous colonies (Figure 1). At 46°C, although both strains grew poorly compared to their growth at 37°C, mutant cultures again produced lower OD readings than *rpoBWT* cultures (Figure 2). However, at this temperature, mutant cultures produced higher CFUs than *rpoBWT*, and by the end of the growth experiment (270 min), mutant cultures had 10-fold higher CFUs than *rpoBWT*. In fact, after 150 min, viable cell counts for *rpoBWT* began to drop, while viable cell counts for the mutant rose. These results further strengthened the argument that the *rpoB58* mutant exhibits higher thermotolerance than *rpoBWT*.

2.2 | Cell Morphological Changes by *rpoB58*

Based on a previous report of the effect of a RpoB substitution (P153L) on cell size (Shiver et al. 2021), we speculated that the discrepancy between OD readings and CFUs we observed in Figure 2 could be due to differences in cell sizes between the *rpoBWT* and *rpoB58* mutant cultures. Microscopic examination confirmed that exponentially grown mutant cells were significantly smaller than *rpoBWT* cells during the exponential phase and did not undergo a drastic size shift from exponential to

stationary phase (Figure S1). We conclude that differences in cell size between *rpoBWT* and mutant at the exponential phase likely contribute to *rpoBWT* cultures providing higher OD readings than mutant cultures without correspondingly increasing CFUs.

A decline in *rpoBWT* viable cell counts beyond the 150-min time point at 46°C (Figure 2B) indicated significant impairments in cellular processes. However, while the *rpoBWT* culture's CFU declined, the OD continued to rise during the same period. We considered that the arrest of cell division, resulting in filamentation, could cause OD to rise without actually increasing CFU. Indeed, growth of *E. coli* cultures at high temperature ($\geq 45^\circ\text{C}$) has been reported to cause cell filamentation (Hoffman and Frank 1963; van Derlinden et al. 2008). We employed scanning electron microscopy to capture finer morphological details (Figure 3). At 37°C, *rpoBWT* and mutant cells from stationary phase culture have almost identical dimensions, appearing as short, ovoidal rods. However, at 46°C, cell length increased dramatically for both strains, with *rpoBWT* cells appearing much larger than mutant cells and more frequently lacking visible septa. These results demonstrate that in comparison to *rpoB58* mutant cells, *rpoBWT* cells at 46°C are under considerably greater stress and frequently stop dividing.

2.3 | Survival Under Extreme Heat Shock and Without Chaperones

The thermotolerance phenotype of the *rpoB58* mutant was further probed by two means: first, survival under hyper-thermal shock of 55°C, and second, survival without molecular

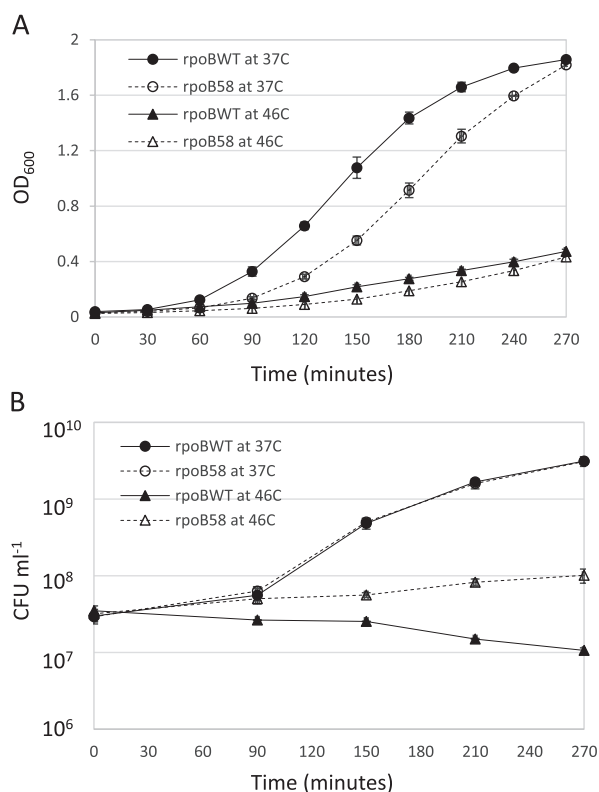


FIGURE 2 | Growth of *rpoBWT* and *rpoB58* cultures in lysogenic broth. (A) Growth curves of bacterial cultures grown at 37°C or 46°C. Cells were grown overnight at 37°C. The next day, overnight cultures were diluted 1:100 in flasks containing fresh lysogenic broth and growth was resumed at 37°C and 46°C. OD₆₀₀ was measured from bacterial samples withdrawn every 30 min. (B) Colony forming units were determined by plating diluted samples withdrawn at 0, 90, 150, 210, and 270 min on lysogenic broth agar. Plates were incubated at 37°C for 24 h. Error bars representing standard deviation are shown ($n = 4$, 37°C; $n = 3$, 46°C).

chaperones. When 37°C-grown cells were shocked at 55°C, *rpoBWT* CFU had dropped over three logs by the 10-min time point, while mutant CFU had dropped by only about 2-fold (Figure S2). The data showed that the mutant had considerably higher tolerance to extreme heat shock than the *rpoBWT*.

The thermotolerance phenotypes of the *rpoB58* mutant raised the possibility of the mutant being less dependent than *rpoBWT* on certain molecular chaperones at elevated growth temperatures. Molecular chaperones play a crucial role in protein folding and degradation (Tomoyasu et al. 2001). DnaK, an ATP-dependent chaperone of the HSP70 class, acts in concert with its co-chaperone DnaJ and the nucleotide exchange factor GrpE to assist in protein folding (Mayer 2021). While GrpE is essential, DnaK and DnaJ are only conditionally essential due to the presence of partially redundant chaperones in *E. coli* cells, such that the absence of DnaKJ causes a temperature-sensitive growth phenotype (Paek and Walker 1987; Sell et al. 1990). The deletion of *dnaJ* alone from the *rpoBWT* background had no effect on growth at 30°C (Figure 4A), but single colony formation was acutely inhibited at 37°C and 42°C (Figure 4B,C). In contrast

to *rpoBWT*, the deletion of *dnaJ* produced no apparent growth defect of the *rpoB58* mutant at 37°C and 42°C (Figure 4B,C). It is noteworthy that while *rpoB58* overcomes the temperature-sensitive growth defect of Δ *dnaJ*, it is unable to restore lambda phage sensitivity without DnaJ (Figure S3), which is required for lambda phage DNA replication (Sunshine et al. 1977; Liberek et al. 1988).

Sell et al. (1990) reported that strains carrying *dnaJ* insertion mutations became filamentous at 42°C. Since *rpoB58* overcomes Δ *dnaJ*'s temperature-dependent growth defect, we also inquired whether it could reverse the cell morphology and/or division defects of Δ *dnaJ*. At 30°C, both *rpoBWT* Δ *dnaJ* and *rpoB58* Δ *dnaJ* strains produced morphologically similar cells at stationary phase, with occasional long filaments observed in both cultures (Figure 5). When grown at a nonpermissive temperature of 42°C, significantly more filamentous and septa-free cells were visible in *rpoBWT* cultures than in *rpoB58* cultures. Thus, the reversal of the temperature-sensitive growth defect of Δ *dnaJ* by *rpoB58* was accompanied by a reversal of the cell division defect at the nonpermissive temperature.

DnaK, unlike its co-chaperone DnaJ, is thought to be a central hub of the protein interaction network (Calloni et al. 2012). We tested whether *rpoB58* could still overcome the more severe growth defect caused by the simultaneous absence of both DnaKJ. For this, we employed a recombinant construct in which expression of the chromosomal *dnaKJ* genes was placed under the IPTG-inducible *lac* promoter (Tomoyasu et al. 1998). The recombinant construct was transduced into *rpoBWT* and *rpoB58* backgrounds using the linked Cm^r marker. All Cm^r transductants grew well in the presence of IPTG (Figure 6D–F). However, in the absence of IPTG, *rpoBWT* cells struggled to form single colonies even at 30°C (Figure 6A) and failed to form single colonies at 37°C and 42°C (Figure 6B,C). In contrast, the absence of IPTG had no effect on the mutant's ability to form single colonies at all three temperatures tested (Figure 6A–C). Thus, *rpoB58* suppresses growth defects resulting from the simultaneous absence of the two major cytoplasmic molecular chaperones, DnaKJ.

2.4 | Survival Without Major Proteases

In addition to chaperones, proteases represent a complementary branch of protein homeostasis. Given *rpoB58*'s ability to overcome deficiency of the molecular chaperones DnaKJ, we asked whether it can also overcome deficiency of major cytoplasmic proteases. To test this, we introduced null alleles of the genes coding for Lon or ClpP protease in *rpoBWT* and *rpoB58* backgrounds and tested growth at a nonpermissive temperature. Consistent with previous reports (Maurizi et al. 1985, 1990), individual null mutations in *lon* or *clpP* did not produce any significant growth defect at 37°C (Figure 7A). However, at 45°C, the *rpoBWT* strain carrying a null allele of *lon* or *clpP* showed a significant growth defect (Figure 7B). In contrast, the absence of Lon or ClpP in the *rpoB58* background produced no obvious growth defect at 45°C (Figure 7B). These data showed that *rpoB58* overcomes significant perturbations

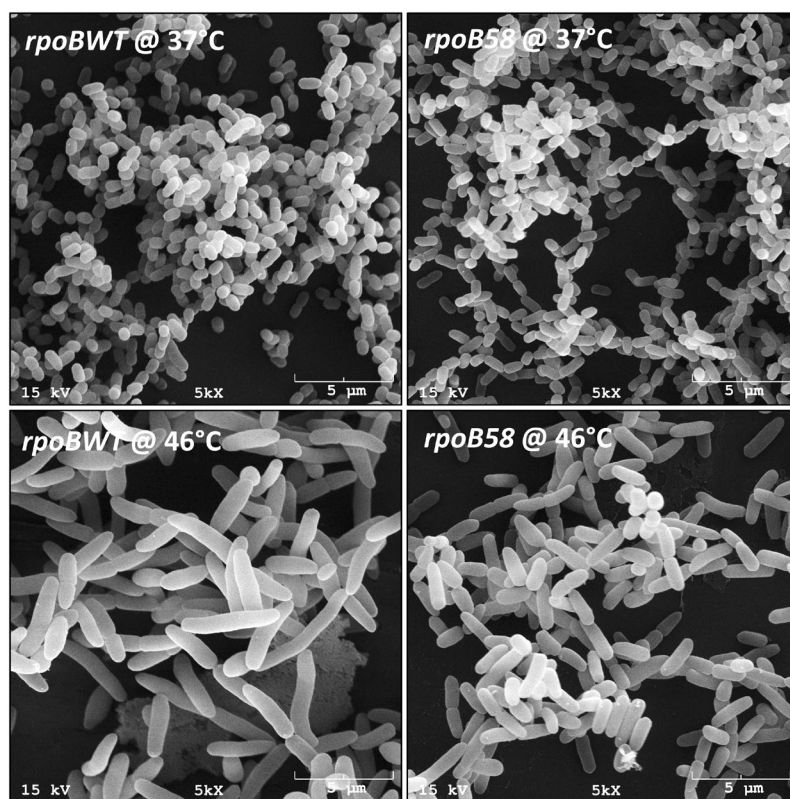


FIGURE 3 | Scanning electron microscopic examination of *rpoBWT* and *rpoB58* cells. Cells were imaged from stationary phase cultures (approximately 5 h after the start of the growth experiment) grown at 37°C (top) or 46°C (bottom). All cultures were grown in lysogenic broth. Average cell dimensions and standard deviations were: *rpoBWT* at 37°C, length = $1.06 \pm 0.2 \mu\text{m}$, width = $0.58 \pm 0.04 \mu\text{m}$ ($n = 105$ cells); *rpoBWT* at 46°C, $3.44 \pm 0.92 \mu\text{m}$, width = $0.92 \pm 0.09 \mu\text{m}$ ($n = 80$ cells); *rpoB58* at 37°C, length = $1.1 \pm 0.18 \mu\text{m}$, width, $0.54 \pm 0.03 \mu\text{m}$ ($n = 105$ cells); *rpoB58* at 46°C, length = $2.65 \pm 0.63 \mu\text{m}$, width, $0.76 \pm 0.08 \mu\text{m}$ ($n = 80$ cells).

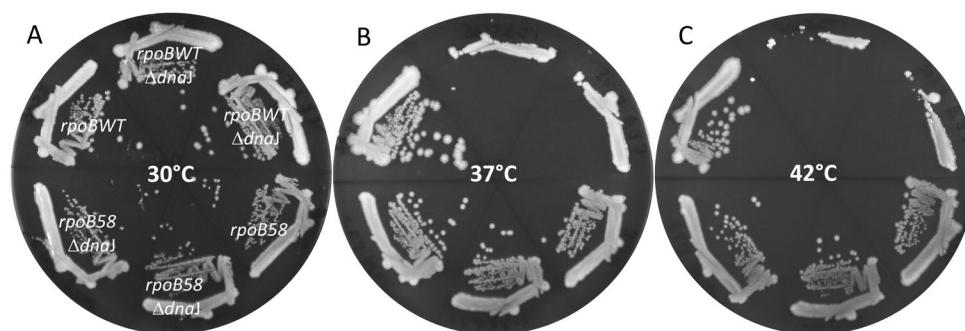


FIGURE 4 | Effects of ΔdnaJ on growth at various temperatures. (A–C) *rpoBWT*, *rpoB58*, and their ΔdnaJ derivatives were tested for their abilities to form single colonies at 30°C (A), 37°C (B), and 42°C (C) on lysogenic broth agar plates for 24 h. Two independent colonies of ΔdnaJ derivatives of *rpoBWT* and *rpoB58* were tested.

in cytoplasmic protein homeostasis caused by the absence of major proteases.

If *rpoB58* influences protein homeostasis more broadly, it may also suppress perturbations occurring outside the cytoplasm. To test this, we deleted *degP*, which codes for a major periplasmic protease (Strauch et al. 1989). As expected, deletion of *degP* from the *rpoBWT* background caused an acute growth defect at 42°C (Figure 7C,D). While deletion of *degP* also caused a growth defect at 42°C in the *rpoB58* background, this defect was far less severe than that observed in the *rpoBWT* background (Figure 7C,D). From this we surmise that

rpoB58 globally influences protein homeostasis to suppress perturbations in both the cytoplasmic and extra-cytoplasmic compartments.

2.5 | Does SR and Its Pre-Activation Contribute to Thermotolerance?

Being pleiotropic, *rpoB58* affects the expression of hundreds of genes, many of which are involved in distinct stress mitigation pathways (Cho and Misra 2021). Its gene expression profile resembles a major aspect of SR: the expression of many

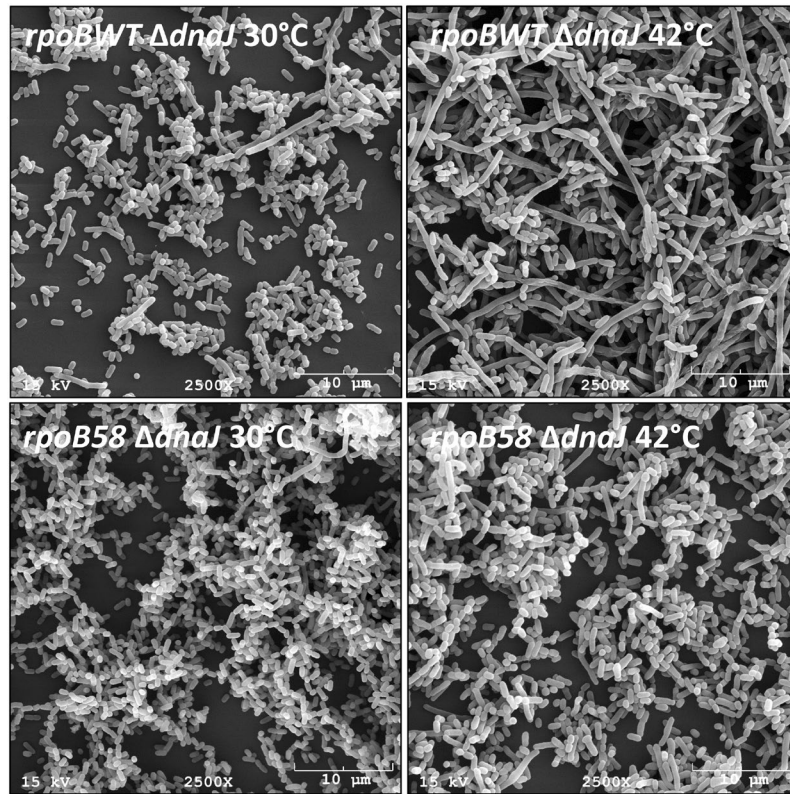


FIGURE 5 | Scanning electron microscopic examination of *rpoBWT ΔdnaJ* and *rpoB58 ΔdnaJ* cells. Cultures were grown at 30°C (left) or 42°C (right) to the stationary phase. Cell size measurements were not carried due to a remarkable heterogeneity in cell morphology of *rpoBWT ΔdnaJ* cultures grown at 42°C.

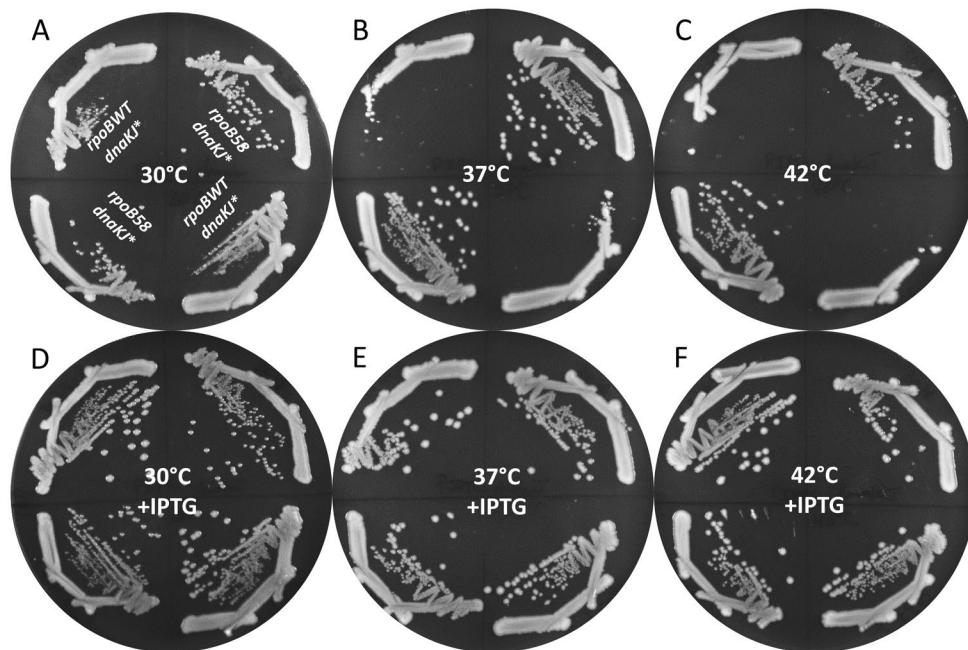


FIGURE 6 | Growth of *rpoBWT* and *rpoB58* strains with and without DnaKJ. Expression of the native chromosomally-localized *dnaKJ* genes (denoted as *dnaKJ**) was controlled by an IPTG-controlled *lac* promoter. Bacterial growth was recorded on lysogenic broth agar after incubating petri plates at 30°C, 37°C, and 42°C for 24 h. Expression of *dnaKJ* was either not induced (A-C) or induced by growth on lysogenic broth agar medium supplemented with 100 μM IPTG (D-F).

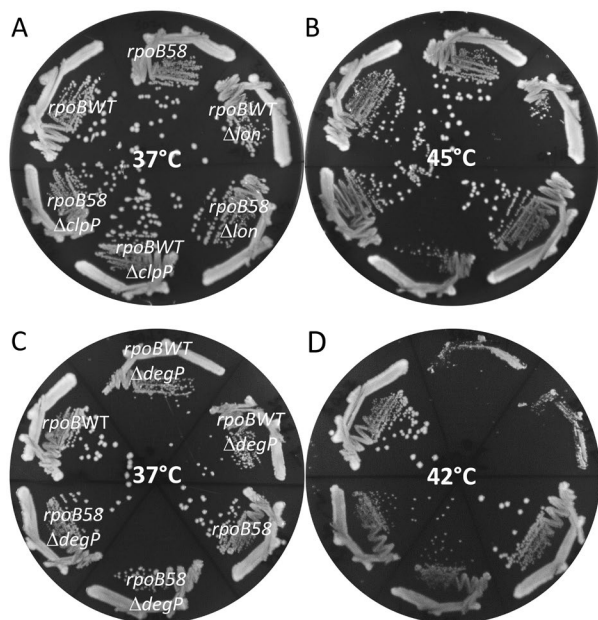


FIGURE 7 | Effects of *lon*, *clpP*, and *degP* deletions on growth of *rpoBWT* and *rpoB58* strains. Effects of *lon* or *clpP* deletion (A–B) or *degP* deletion (C–D) on bacterial growth. Bacterial growth was recorded on lysogenic broth agar after incubating petri plates at temperatures as shown for 24 h. Two independent colonies of $\Delta degP$ derivatives of *rpoBWT* and *rpoB58* were tested.

genes involved in protein synthesis is low, while the expression of genes involved in amino acid biosynthesis is elevated (Sanchez-Vazquez et al. 2019; Cho and Misra 2021). Moreover, *rpoB58* is thought to transform the RNA polymerase into a “stringent” polymerase, since it overcomes amino acid auxotrophy without DksA (Cho and Misra 2021), a critical SR determinant required for the synthesis of certain amino acids (Ross et al. 2016). Based on these similarities, we hypothesized that the activation of the stringent-like response (S-LR) pathway by *rpoB58* is responsible for overcoming perturbations in protein homeostasis.

We tested this hypothesis by comparing thermotolerance of four *rpoBWT* strains: *relA*⁺ *spoT*⁺, $\Delta relA$ *spoT*⁺, $\Delta relA$ $\Delta spoT$, and *relA*^{QUAD}. While the $\Delta relA$ *spoT*⁺ strain is not expected to produce any (p)ppGpp, the hyperactive *relA*^{QUAD} allele constitutively increases the basal (p)ppGpp level compared to the *relA*⁺ allele (Sinha and Winther 2021). We initially chose 45°C as the test temperature because we had previously determined it to be the upper limit for growth of the stringent-defective *relA1* strain (Figure 1). At 45°C, *relA*⁺ *spoT*⁺, $\Delta relA$ *spoT*⁺, and *relA*^{QUAD} strains formed similar-sized colonies (Figure 8B), but colonies formed by the $\Delta relA$ $\Delta spoT$ strain were visibly smaller than other strains (Figure 8B). When the incubation temperature was raised to 45.5°C, growth of the $\Delta relA$ $\Delta spoT$ strain was severely impaired, while $\Delta relA$ *spoT*⁺ struggled to form homogeneous single colonies (Figure 8C). In contrast,

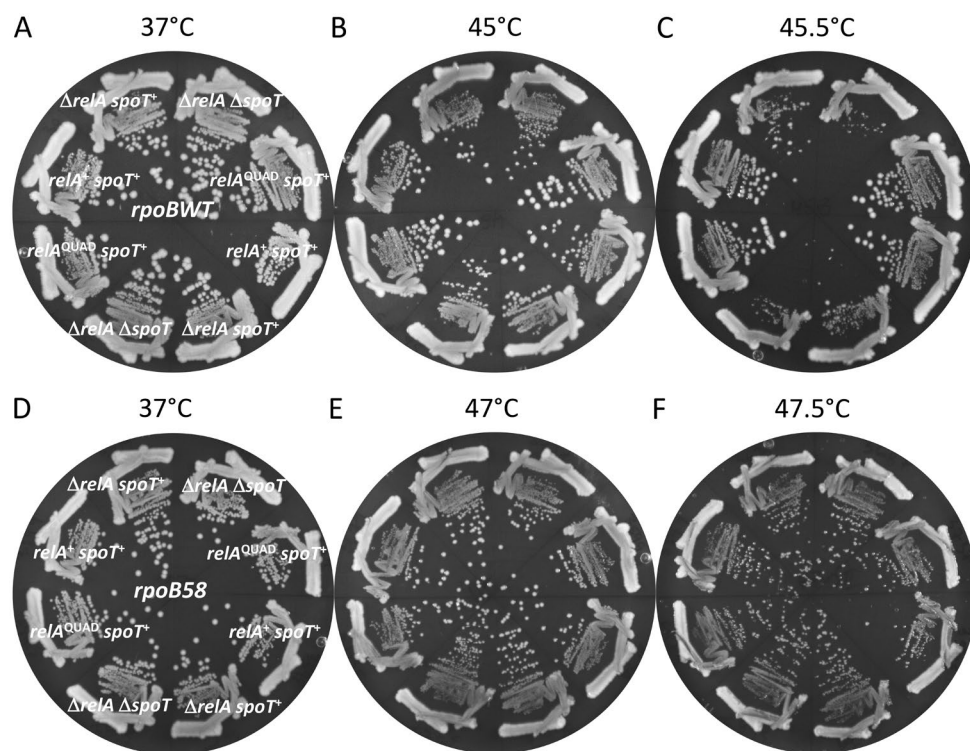


FIGURE 8 | Effects of different *relA* alleles on thermotolerance. (A–C) Bacterial growth of *rpoBWT* strains carrying *relA*⁺, $\Delta relA$, $\Delta relA$ $\Delta spoT$, or *relA*^{QUAD} alleles was recorded on lysogenic broth agar after incubating petri plates at the threshold temperature of 45°C (B) or above (C) for 24 h. Colonies for this test were taken from a plate grown at 37°C (A), where all strains grew normally. (D–F) The ability of *rpoB58* strains carrying *relA*⁺, $\Delta relA$, $\Delta relA$ $\Delta spoT$, or *relA*^{QUAD} alleles to grow at the threshold temperature of 47°C (E) or above (F) was tested on lysogenic broth agar after incubating petri plates for 24 h. Colonies for this test were taken from a plate grown at 37°C (A) or (D), where all strains grew normally.

relA⁺ *spoT*⁺ and *relA*^{QUAD} strains grew normally (Figure 8C). The highest temperatures where *relA*⁺ *spoT*⁺ and *relA*^{QUAD} strains could still form single colonies were determined to be 46°C and 46.1°C, respectively (Figure S4). These data showed that SR proficiency and SR pre-activation contribute to thermotolerance.

In parallel, we tested the effect of *relA*/*spoT*-mediated SR on thermotolerance conferred by *rpoB58* (Figure 8D–F). Regardless of the nature of the *relA* and *spoT* alleles, all *rpoB58* strains were able to grow and formed single colonies at 47°C (Figure 8E), the same threshold temperature of growth as the *rpoB58 relA1* strain (Figure 1). These strains were also able to grow and form single colonies even at 47.5°C (Figure 8F), but not at 48°C (Figure S4). One reason for the apparent indifference of the *rpoB58* strain to various *relA* and *spoT* alleles could be that the mutant polymerase, unlike the wild type polymerase, does not bind to or respond to (p)ppGpp. We tested this possibility by examining growth on M63-glucose minimal medium (M63-glu) containing 100 µg mL⁻¹ each of serine, methionine, and glycine (SMG) that permits growth based on the strength of the SR-mounting capability (Uzan and Danchin 1976). The data showed that growth of the *rpoB58* strain, like that of the *rpoBWT* strain, on M63-glu + SMG medium corresponded to the cell's capacity to produce (p)ppGpp, with slowest and fastest growth observed in the *relA1* and *relA*^{QUAD} backgrounds, respectively (Figure S5). From this, we conclude that the RpoB58 polymerase still responds to the level of (p)ppGpp in the cell for overcoming the classic SR phenotype, but, unlike the RpoBWT polymerase, it is not dependent on (p)ppGpp for thermotolerance.

2.6 | Role of SR in Chaperone Independence

Because SR proficiency and SR pre-activation contribute to thermotolerance in the *rpoBWT* background (Figure 8), we tested the effects of *relA*⁺ and *relA*^{QUAD} in a *rpoBWT* background lacking DnaJ at various growth temperatures (Figure 9A–C; data for the stringent-defective *relA1* strain are shown in Figure 4A–C). At 37°C, both *relA*⁺ and *relA*^{QUAD} strains formed single colonies without DnaJ, although *relA*^{QUAD} facilitated better growth than *relA*⁺ (Figure 9A). When the incubation temperature was raised to 40°C, the *relA*⁺ Δ *dnaJ* strain began to struggle, while the *relA*^{QUAD} Δ *dnaJ* strain continued to form homogeneous single colonies (Figure 9B). At 42°C, both strains failed to form single colonies (Figure 9C). It is worth noting that a *relA1* Δ *dnaJ* strain failed to form single colonies even at 37°C (Figure 4B). Together, these results show that both SR and its pre-activation by *relA*^{QUAD} contribute to chaperone independence, albeit not as robustly as that exhibited by *rpoB58* (Figure 4B,C).

While *rpoB58*-mediated suppression of Δ *dnaJ*'s growth defect is independent of a functional *relA* gene (Figure 4A–C), we asked whether growth suppression by *rpoB58* would prevail in the complete absence of (p)ppGpp synthetase activity in a Δ *relA* Δ *spoT* background. Due to a well-known multi-amino acid auxotrophy of a (p)ppGpp⁰ strain (Xiao et al. 1991), *rpoBWT* and *rpoBWT* Δ *dnaJ* strains lacking *relA* and *spoT* failed to grow on a glucose minimal medium not supplemented with amino acids (Figure 9D). In contrast, the presence of *rpoB58* supported some growth without *relA* and *spoT* (Figure 9D), reflecting the “stringent” nature of the RpoB58 polymerase. Interestingly, regardless of the (p)ppGpp synthesizing capacity of the background,

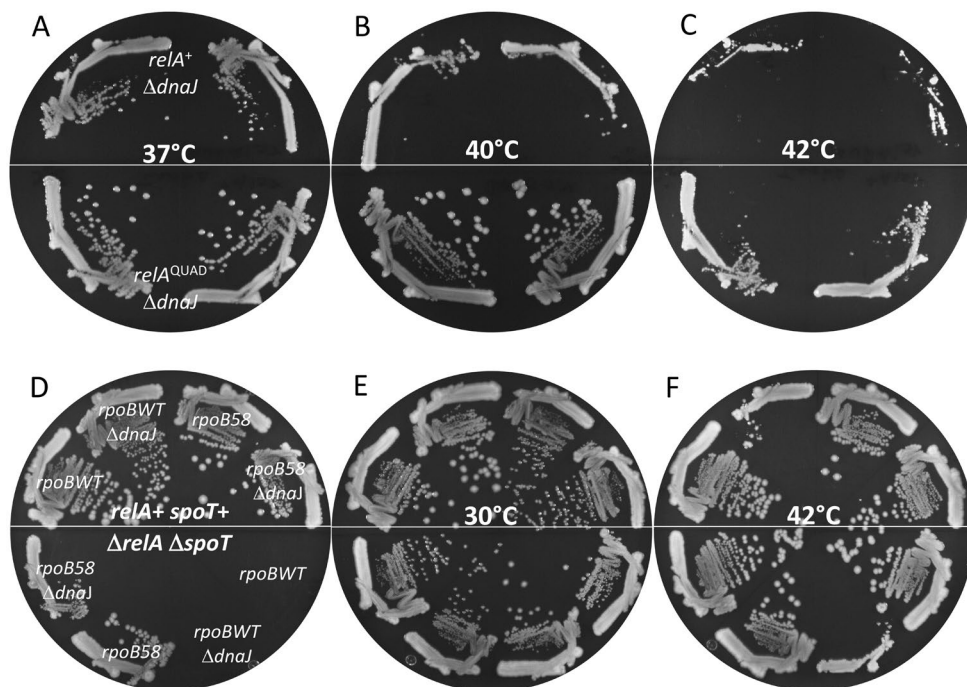


FIGURE 9 | Effects of *relA*⁺, *relA*^{QUAD}, and Δ *relA* Δ *spoT* on growth with or without DnaJ at various temperatures. (A–C) Bacterial growth was recorded on lysogenic broth agar after incubating petri plates at 37°C, 40°C, and 42°C for 24 h. Two independent colonies of each strain were tested. (D–F) Bacterial growth was recorded on M63-minimal medium (D) or lysogenic broth agar (E, F) after incubating petri plates at 37°C (D), 30°C (E), and 42°C (F) for 24 h. Strains used in the top half of the plate contain functional *relA* and *spoT* alleles, while those in the bottom half of the plate are deleted for these two genes.

rpoB58 readily overcame $\Delta dnaJ$'s growth defect at a nonpermissive temperature (Figure 9F).

2.7 | DksA Is Required for Chaperone Independence by *rpoB58*

As stated above, DksA is a critical stringent factor that facilitates binding of (p)ppGpp to site II on RNAP and acts synergistically with (p)ppGpp to decrease transcription from the rRNA and r-protein genes (Lemke et al. 2011; Ross et al. 2016). Deletion of *dksA* causes a "relaxed" phenotype, including increased transcription from the promoter of the ribosomal gene *rrnB* (Paul et al. 2004). We hypothesized that in a $\Delta dksA$ background, increased transcription of genes involved in protein synthesis would exacerbate the growth defect of a strain lacking DnaKJ. We first confirmed the effect of $\Delta dksA$ on *rrnB* by employing a *rrnB* P1::lacZ construct (Paul et al. 2004). The deletion of *dksA* elevated *rrnB* P1::lacZ expression by 4-fold in the *rpoBWT* background (Figure 10A). In contrast, expression of *rrnB* P1::lacZ was reduced by 40% in the presence of *rpoB58*. Although the introduction of *rpoB58* in a $\Delta dksA$ background lowered expression of *rrnB* P1::lacZ by 30% compared to *rpoBWT* $\Delta dksA$, it remained 3-fold higher than that seen in the *rpoBWT* (*rpoB*⁺ *dksA*⁺) strain. These results show that

rpoB58 constitutively reduces protein synthesis but still becomes partially "relaxed" in the absence of DksA.

Results presented in Figure 10B–D show the effects of the simultaneous absence of *dksA* and *dnaKJ*. In the *rpoBWT* background, the absence of both *dksA* and *dnaKJ* is debilitating and prevents formation of single colonies at and above 35°C. While *rpoB58* still improved growth without DnaKJ at 35°C (Figure 10C), its ability to form single colonies was abolished at 36°C (Figure 10D). Both strains grew normally in the presence of IPTG at 37°C (Figure S6). Note that in the DksA⁺ background, *rpoB58* overcomes growth defects without DnaKJ (minus IPTG) even at 42°C (Figure 6). Together, these data show that even though *rpoB58* triggers S-LR, its ability to effectively overcome perturbation in protein homeostasis without chaperones requires DksA, which is a critical factor in lowering protein synthesis.

2.8 | Slowing Translation Enhances Cell Survival Without a Chaperone

Based on the results obtained above, we hypothesized that lowering protein synthesis may enhance cells' ability to survive without chaperones. We tested this hypothesis by two independent means. First, we directly lowered protein synthesis using sub-lethal dosages of chloramphenicol and then examined whether this allowed growth of *rpoBWT* cells lacking DnaJ at a nonpermissive temperature. Cells were first grown in the presence of a sub-lethal dosage (0.625 or 1.25 $\mu\text{g mL}^{-1}$) of chloramphenicol at 30°C to slow down protein synthesis (Figure 11A). From here, cells were then streaked onto LBA plates containing 0.625, 1.25 $\mu\text{g mL}^{-1}$, or no chloramphenicol and incubated at 38.5°C (Figure 11B–D), which

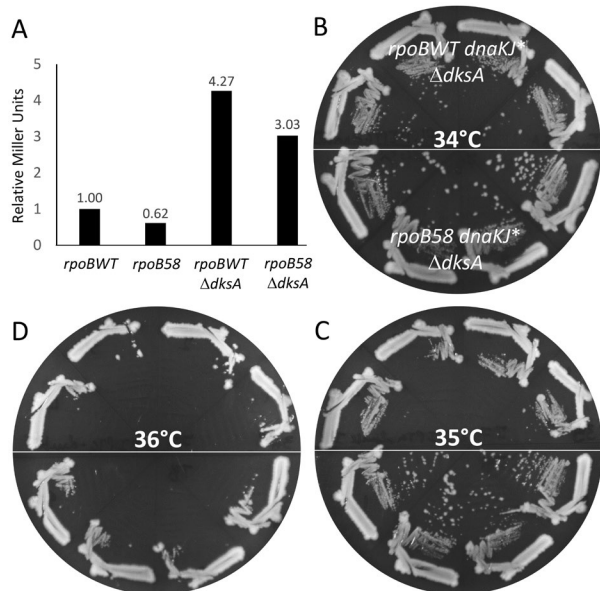


FIGURE 10 | Effects of *rpoB58* and $\Delta dksA$ on *rrnB* P1::Z expression and their ability to overcome DnaKJ deficiency. (A) Beta-galactosidase assays were conducted from cultures of *rpoBWT*, *rpoB58*, *rpoBWT* $\Delta dksA$, and *rpoB58* $\Delta dksA$ strains grown at 37°C in lysogenic broth. Assays were carried out from three independent cultures in duplicate. Relative beta-galactosidase activity as Miller units are shown. *p*-values of beta-galactosidase activity differences between *rpoBWT* and *rpoB58*, as well as between *rpoBWT* $\Delta dksA$ and *rpoB58* $\Delta dksA$ strains, were less than 0.001. (B–D) *rpoBWT* and *rpoB58* strains deleted for *dksA* and expressing the native chromosomally-localized IPTG-inducible *dnaKJ* genes (denoted as *dnaKJ*^{*}) were tested for growth either without IPTG at 34°C, 35°C, and 36°C. Bacterial growth was recorded on lysogenic broth agar after incubating petri plates for 24 h. Four independent colonies of each strain were tested.

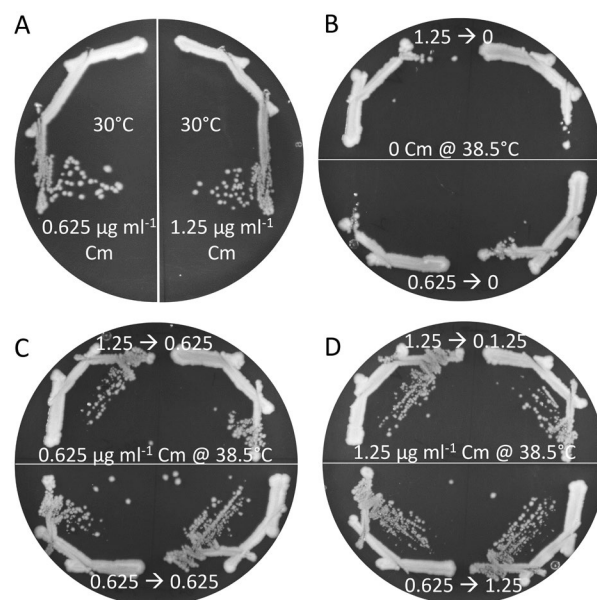


FIGURE 11 | Effects of inhibition of protein synthesis by chloramphenicol on growth without DnaJ. (A) *rpoBWT* $\Delta dnaJ$ cells grown on lysogenic broth agar containing 0.625 or 1.25 $\mu\text{g mL}^{-1}$ of chloramphenicol for 36 h at 30°C. (B–D) Colonies from chloramphenicol plates grown at 30°C (A) were then streaked on plates containing either no chloramphenicol (B), 0.625 $\mu\text{g mL}^{-1}$ (C), and 1.25 $\mu\text{g mL}^{-1}$ (D) of chloramphenicol and incubated for 24 h at 38.5°C.

is above the nonpermissive temperature for single colony formation of *rpoBWT ΔdnaJ* cells (Figure 4B). When chloramphenicol was present in the medium, *rpoBWT ΔdnaJ* cells displayed improved growth and formed single colonies at a nonpermissive temperature (Figure 11C,D). This supported the hypothesis that a reduction in protein synthesis could improve survival at nonpermissive temperatures in the absence of chaperones.

Second, we employed minimal medium to slow down the growth rate and protein synthesizing capacity (Tao et al. 1999; Li et al. 2014). We used *rrnB* P1::lacZ to indirectly assess the effects of growth medium on protein synthesizing capacity. Growth in the M63-glu minimal medium resulted in 40% lower *rrnB* P1::lacZ activity than from cultures grown in LB, and *rpoB58* continued to repress *rrnB* P1::lacZ activity regardless of the medium composition (Figure 12A). To determine the specificity of the effect of *rpoB58* on *rrnB* P1::lacZ, we used a control promoter fusion construct, *lacUV5::lacZ* (Paul et al. 2004). Unlike the negative effect of *rpoB58* on *rrnB* P1::lacZ, the activity of *lacUV5::lacZ* was identical in both *rpoBWT* and *rpoB58* backgrounds, regardless of the growth medium used (Figure 12A). We then examined whether the reduction of protein synthesis by growth on minimal medium could overcome temperature defects in the absence of DnaKJ. *rpoBWT* and *rpoB58* strains with

or without DnaKJ were first grown at a permissive growth temperature of 30°C, then streaked onto LBA and M63-glu plates and incubated at nonpermissive temperatures of 37°C and 42°C (Figure 12B,C). As expected, without DnaKJ, *rpoBWT* cells grew extremely poorly when streaked at nonpermissive temperatures on LBA (Figure 12B) but grew well and formed single colonies on M63-glu plates at these temperatures (Figure 12C). Together, these results support our hypothesis that growth defects caused by a collapse in protein homeostasis can be partially overcome by lowering the cell's capacity for protein synthesis.

3 | Discussion

Growth of *E. coli* strains incubated at 44°C in minimal medium is restricted principally due to the inactivation of the first enzyme (MetA) of methionine biosynthesis (Ron and Davis 1971). However, what restricts high-temperature growth of *E. coli* in rich medium is not clear. The maximum growth temperature of *E. coli* incubated in rich medium is reported to be between 44.5°C and 46°C (Hoffman and Frank 1963; van Derlinden et al. 2008; Rudolph et al. 2010). For the *E. coli* strain background (MC4100) used in this study, the maximum growth temperature in a rich medium was determined to be between 45°C and 46°C. The presence of a mutant *rpoB58* allele increased this limit to between 47°C and 48°C. The inhibition of *E. coli* growth at temperatures higher than 46°C is likely due to debilitating effects on cell structure–function, with a collapse in protein homeostasis being one of the major factors. Indeed, null mutations in genes central to the maintenance of protein homeostasis, such as those coding for molecular chaperones and proteases, significantly lower the maximum growth temperature and are classically linked to growth defects at temperatures between 37°C and 42°C (Paek and Walker 1987; Strauch et al. 1989; Sell et al. 1990; Ueguchi et al. 1995).

At an extreme high temperature between 46°C and 47°C, null mutations in 51 *E. coli* genes were shown to become essential for survival (Murata et al. 2011). In addition to some of the obvious candidate genes, such as chaperone-coding *dnaK* and *dnaJ* and protease-coding *degP*, the screen identified genes involved in diverse cellular functions, including outer membrane biogenesis, cell division, and energy metabolism (Murata et al. 2011). Due to their essential nature, none of the genes directly involved in protein synthesis were identified in the screen, but the authors noted reduced expression of 23 genes that participate in protein synthesis. Based on this observation, they concluded that down-regulation of ribosomal genes is one of the possible strategies for survival at extremely high growth temperatures.

Both the activated SR (Sanchez-Vazquez et al. 2019) and the S-LR activated by *rpoB58* (Cho and Misra 2021) alter expression of hundreds of genes, with the largest number of negatively affected genes being those involved in protein synthesis. By lowering protein synthesis, activated SR and S-LR are expected to lower stress when protein homeostasis is destabilized at high incubation temperatures or in the absence of chaperones (DnaKJ) and proteases (ClpP, Lon, and DegP). Because *rpoB58* was isolated in the SR-defective *relA1* background, its effect on protein synthesis is expected to be largely due to decreased transcription of genes

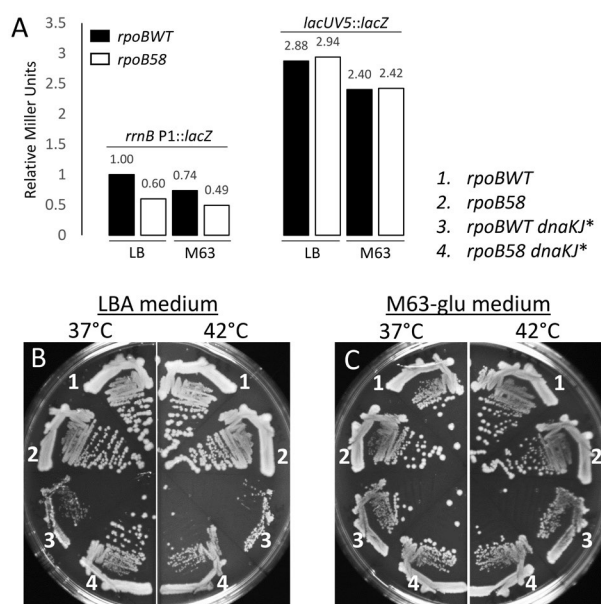


FIGURE 12 | Effects of growth media on *rrnB* P1::lacZ expression and cell's ability to overcome DnaJ deficiency. (A) Beta-galactosidase assays were conducted from cultures of *rpoBWT* and *rpoB58* strains grown overnight at 37°C in lysogenic broth. Assays were carried out from three independent cultures in duplicate. Relative beta-galactosidase activity as Miller units are shown. For *rrnB* P1::lacZ, *p*-values of beta-galactosidase activity differences between *rpoBWT* and *rpoB58* strains were less than 0.001. For *lacUV5*::lacZ, no significant differences in beta-galactosidase activities were recorded between *rpoBWT* and *rpoB58* strains. (B, C) *rpoBWT* and *rpoB58* strains with (labelled 1 and 2, respectively) or without DnaKJ (labelled 3 and 4, respectively) were tested for growth on lysogenic broth agar (B) or M63-glu minimal (C) medium at temperatures shown for 24 h. Strains without DnaKJ were grown in the absence of IPTG so that the expression of the chromosomally-localized *dnaKJ** would not be induced.

involved in protein synthesis (Cho and Misra 2021). In contrast, the *relA*^{QUAD}-mediated constitutive activation of SR is expected to act both at the level of transcription through (p)ppGpp-binding to RNAP (Ross et al. 2016) as well as through a direct effect of elevated (p)ppGpp levels on translation (Milon et al. 2006).

Unlike *rpoB58* (Figure 1) and *relA*^{QUAD} (Figure 8), *relA*⁺ does not appreciably reverse the growth defect at 46°C (Figure 8) or the defect resulting from the absence of DnaJ at temperatures of 40°C or higher (Figure 9). This suggests that pre-activation of SR (*relA*^{QUAD}) or S-LR (*rpoB58*) is more beneficial to cell survival under thermal stress conditions than simply being capable of activating SR, as in the *relA*⁺ background. Nevertheless, growth defects at 46°C or without DnaJ at temperatures ≥ 40°C are milder in the *relA*⁺ background than in the *relA1* background (Figures 4 and 9), indicating that having the ability to mount SR confers some benefit over a background unable to do so. The ability of *rpoB58* to withstand an extreme heat shock of 55°C without pre-adaptation to higher temperatures once again points to *rpoB58* cells having a preexisting physiological state conferring thermotolerance. Although we do not currently know the precise nature of this physiological state, SR appears to play an integral role.

A feedback regulatory loop exists to balance the production of cytoplasmic chaperones and proteases in response to protein misfolding (Straus et al. 1990). Similarly, misfolding and misassembly of envelope proteins trigger regulatory systems to lower envelope stress (Danese and Silhavy 1997; Ades et al. 1999; Gerken et al. 2010). One such regulatory system, controlled by σ^E , not only upregulates expression of envelope chaperones and proteases in response to protein misfolding, but also lowers the synthesis of selected outer membrane proteins through small regulatory RNAs (Johansen et al. 2006) to stem their flow to the compromised protein assembly environment. In contrast to these well-studied regulatory systems, relatively little is known about the existence of a feedback regulatory loop in *E. coli* that regulates protein synthesis in response to protein misfolding in the cytoplasm. One study showed that plasmid-mediated overexpression of Hsp33 in a DnaKJ-deficient background rescued growth by triggering degradation of EF-Tu, an essential elongation factor of protein synthesis, by the Lon protease, leading to reduced protein aggregation (Bruehl et al. 2012). The same study also showed that overexpression of HipA, the glutamyl-tRNA synthetase-targeting toxin component of the HipAB toxin-antitoxin module (Kaspy et al. 2015), partially rescued growth in a background deficient in the major cytoplasmic chaperones. HipA-mediated inactivation (phosphorylation) of glutamyl-tRNA synthetase leads to the accumulation of uncharged tRNA^{Glu}, stalling protein synthesis and triggering SR (Kaspy et al. 2015). Another study showed that late steps of ribosome assembly become defective at 45°C or at lower temperatures without DnaKJ (René and Alix 2010). These authors suggested that the accumulation of misfolded proteins at an extremely high growth temperature would sequester chaperones like DnaK, thus halting the late steps of ribosome assembly (René and Alix 2010). Slowing translation rate and mild inhibition of protein synthesis have also been linked to improvement in protein folding (Siller et al. 2010; Meriin et al. 2012). These examples clearly demonstrate that cells can be rescued from the

collapse of protein homeostasis in the cytoplasm by lowering their protein synthesizing capacity or rate.

The direct and indirect roles of SR in controlling protein synthesis in response to various stresses make it a good candidate for feedback regulation of protein synthesis in response to disruption of protein homeostasis. In fact, in *Bacillus subtilis*, it was shown that (p)ppGpp-mediated inhibition of protein synthesis is intimately involved in the heat shock response and the development of thermotolerance (Schäfer et al. 2020). The work carried out here similarly links (p)ppGpp-dependent SR and (p)ppGpp-independent S-LR to protein homeostasis and thermotolerance (Figure 13). The following observations support this assertion: (a) SR-proficient *relA*⁺ and *relA*^{QUAD} cells displayed greater thermotolerance at temperatures above 45°C and without DnaJ at temperatures ≥ 40°C than SR-defective *relA1* and $\Delta relA \Delta spoT$ cells, (b) *rpoB58*, which constitutively triggers S-LR, permitted cell growth between 47°C and 48°C, in the absence of DnaJ, DnaKJ, or DegP at 42°C, and at 45°C

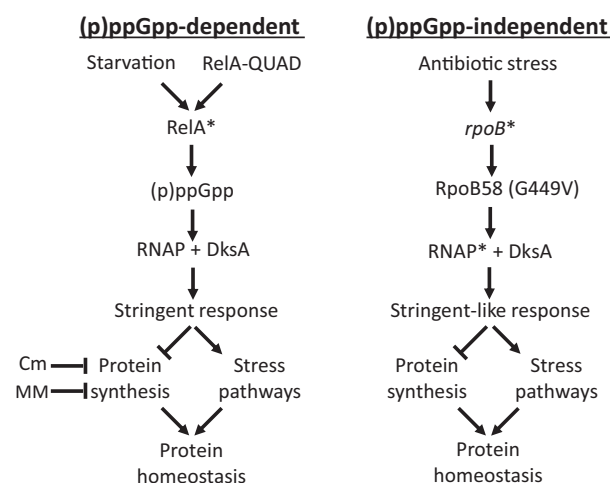


FIGURE 13 | A model showing the potential influence of (p)ppGpp-dependent and (p)ppGpp independent pathways on protein homeostasis. The (p)ppGpp-dependent pathway involves (p)ppGpp-synthetase, RelA, whose activity increases (denoted by RelA* in the model) during starvation stress or by a constitutive mutation, *relA*^{QUAD}. The high levels of (p)ppGpp trigger a stringent response upon binding to RNA polymerase (RNAP). The two main outcomes of this response are to lower macromolecular, including protein, synthesis, and to stimulate various stress-mitigation pathways. Data from this work suggest that a secondary effect of these two outcomes is to improve the cell's protein homeostasis capacity. A slowdown in protein synthesis either by the addition of chloramphenicol (Cm) or growth in minimal medium (MM) also improves the cell's homeostasis capacity. The (p)ppGpp-independent pathway begins with mutations in the polymerase subunit coding genes, such as *rpoB* (denoted by *rpoB** in the model), which can be isolated through various selections, including exposure to antibiotics (Cho and Misra 2021). The altered RpoB protein, that is, RpoB58, constitutively transforms the RNAP complex into a stringent-like state (denoted by RNAP* in the model), which the wild type RNAP normally only attains transiently upon binding to (p)ppGpp. The RNA-seq data have shown that the (p)ppGpp-independent pathway triggers the same two overall outcomes as those triggered by the (p)ppGpp-dependent pathway (Cho and Misra 2021), including improvement of the cell's protein homeostasis capacity. DksA plays a critical role in both pathways.

without Lon and ClpP proteases, and (c) in the absence of integral SR factor DksA, which together with (p)ppGpp lowers the expression of the rRNA and r-protein genes, the ability of *rpoB58* to overcome DnaKJ deficiency was dramatically curtailed. Two additional data from this work support the model presented in Figure 13, which shows that lowering the protein synthesis capacity enables cells to rebalance protein homeostasis in the absence of chaperones. First, a direct inhibition of protein synthesis by sub-lethal dosages of chloramphenicol improves growth of the *dnaJ* mutant at a nonpermissive temperature (Figure 11). Second, growth in minimal medium, known to slow down growth rate and protein synthesizing capacity (Tao et al. 1999; Li et al. 2014), also reverses the growth defect without DnaKJ at nonpermissive temperatures (Figure 12). Thus, the positive effects of *rpoB58* on protein homeostasis are mimicked by *rpoBWT* under growth conditions that lower protein synthesis. Interestingly, slow growth on minimal medium without supplementation of amino acids is known to elevate (p)ppGpp levels (Imholz et al. 2020), which could be one of the driving factors in lowering rRNA expression and protein synthesis. It is worth noting that while the RNAP complex assembled with the RpoB58 mutant protein still responds to (p)ppGpp levels for strictly SR-related phenotypes (Figure S5), the presence of (p)ppGpp is not required for the RpoB58-mediated restoration of protein homeostasis (Figure 9).

Two previous observations also linked SR to thermotolerance. First, it was noted that the *relA1* mutant failed to grow at temperatures above 41°C in a medium without salt (Yang and Ishiguro 2003). This temperature-sensitive growth defect was reversed either by the addition of salt or by the introduction of “stringent” alleles of *rpoB* (Yang and Ishiguro 2003). Second, a genetic screen of null mutations identified *dksA* as one of the genes whose product is required for survival at 47°C (Murata et al. 2011). Given the role of DksA as one of the stringent factors known to lower expression of the rRNA and r-protein genes, it is possible that elevated protein synthesis in the absence of DksA further exacerbates the already strained protein homeostasis at 47°C. It is important to note that while the absence of DksA is detrimental to growth in cells lacking DnaKJ (Figure 10), overexpression of DksA suppresses the temperature-sensitive growth defect of the *dnaK* or *dnaJ* mutant (Kang and Craig 1990), possibly, in part, by lowering protein synthesis. It is worth emphasizing that although SR and S-LR do not activate pathways directly involved in cytoplasmic and envelope protein folding or proteolysis (Sanchez-Vazquez et al. 2019; Cho and Misra 2021), they activate several other stress-mitigating pathways, and it cannot be ruled out that these other pathways may also contribute to lowering stress caused by protein misfolding.

While *rpoB58* overcomes the Δ *dnaJ* growth defect at nonpermissive temperatures (Figure 4), it failed to reverse the lambda-resistant phenotype conferred by Δ *dnaJ* even at a permissive growth temperature (Figure S3). DnaKJ are essential for the initiation of lambda DNA replication (Sunshine et al. 1977; Liberek et al. 1988), and continued resistance to lambda without DnaJ in the *rpoB58* background (Figure 4) shows that certain DnaJ-dependent and highly specialized functions are not restored despite the overall improvement in cellular stress. In contrast to

the lambda phenotype, *rpoB58* was able to reverse the cell filamentation phenotype of Δ *dnaJ* at a nonpermissive temperature (Figure 5). Depletion of GroE has been shown to cause a cell division defect due to misfolding of the important septal stabilizing protein FtsE (Fujiwara and Taguchi 2007). It is possible that DnaKJ also facilitate the folding of a protein or assembly/dis-assembly of a complex critical for cell division (Bukau and Walker 1989). Indeed, these authors showed that overexpression of a core septal protein, FtsZ, overcomes the filamentation phenotype of Δ *dnaK* at 30°C. The ability of *rpoB58* to restore cell division in the absence of DnaJ at 42°C indicates that the error in protein folding or assembly of the affected cell division complex is corrected in this background.

Finally, it is important to emphasize that while in this work we have focused on one “stringent” *rpoB* mutant, *rpoB58*, which was one of four *rpoB* mutants obtained among antibiotic resistant revertants (Cho and Misra 2021), a number of past studies have reported the isolation of “stringent” RNA polymerase mutants through more direct selection strategies designed to enrich for such mutants (Barker et al. 2001; Trautinger and Lloyd 2002; Murphy and Cashel 2003; Szalewska-Palasz et al. 2007). Such mutants have not been screened for thermotolerance, but it is tempting to suggest that a number of these “stringent” RNA polymerase mutants may be able to overcome chaperone deficiency and gain higher thermotolerance by the mechanism outlined in this paper.

4 | Experimental Procedure

Bacterial strains, growth conditions, and genetic methods: Most *Escherichia coli* strains used in this study (Table 1) are derived from RAM1292 (Werner and Misra 2005), a Δ *ara714* variant of MC4100 (Casadaban 1976). Bacterial cultures were grown in lysogenic broth (LB, Difco LB Mix powder) or lysogenic broth agar (LBA, 1.5%; Becton Dickinson). When desired, IPTG (100 μ M), chloramphenicol (12.5, 2.5, 1.25, or 0.625 μ g mL⁻¹), kanamycin (25 μ g mL⁻¹), or tetracycline (10 μ g mL⁻¹) were added to the medium. Minimal medium (M63) was prepared as described previously (Silhavy et al. 1984) and supplemented with 0.2% glucose. In general, bacterial cells in liquid or solid medium were grown by incubating at 37°C. For testing temperature sensitivity, plates streaked with bacterial strains were incubated at various temperatures, ranging from 30°C to 48°C. For growth curves, cultures were grown overnight at 37°C and diluted back in fresh medium at either 37°C or 46°C. Samples were diluted in LB and incubated on LBA plates at 37°C overnight to determine viable cell count.

P1 phage-mediated transductions were carried out as described previously (Silhavy et al. 1984). RAM1292 and its parental strain MC4100 contain a *relA1* allele that renders the strain SR-incompetent (Metzger et al. 1989) thus preventing growth on a glucose minimal medium containing 150 μ g mL⁻¹ serine or 100 μ g mL⁻¹ each of serine, methionine, and glycine (Uzan and Danchin 1976). The *relA1* allele was replaced with *relA*⁺ or *relA*^{QUAD} by P1 transductions using a linked Δ *queF*::Km^r marker. The replacement of *relA1* by *relA*⁺ or *relA*^{QUAD} was confirmed by growth on glucose minimal medium containing 150 μ g mL⁻¹ serine. Additional confirmation of the replacement

TABLE 1 | Bacterial strains used in this study.

Strain name	Key genotype	Source
RAM1292	MC4100 (<i>relA1</i>) Δ <i>ara714</i>	Werner and Misra (2005)
RAM3027	RAM1292 <i>rpoBWT</i> <i>btuB</i> ::Tn10	Cho and Misra (2021)
RAM3028	RAM1292 <i>rpoB58</i> <i>btuB</i> ::Tn10	Cho and Misra (2021)
RAM3049	RAM3027 Δ <i>degP</i> ::Km ^r	This study
RAM3050	RAM3028 Δ <i>degP</i> ::Km ^r	This study
RAM3174	RAM3027 Δ <i>dnaJ</i> ::Km ^r	This study
RMA3175	RAM3028 Δ <i>dnaJ</i> ::Km ^r	This study
RAM3195	RAM3027 P(IPTG) <i>lac</i> :: <i>dnaKJ</i> -Cm ^r	This study
RAM3197	RAM3028 P(IPTG) <i>lac</i> :: <i>dnaKJ</i> -Cm ^r	This study
RAM3231	RAM3027 Δ <i>dkA</i> ::scar P(IPTG) <i>lac</i> :: <i>dnaKJ</i> -Cm ^r	This study
RAM3232	RAM3028 Δ <i>dkA</i> ::scar P(IPTG) <i>lac</i> :: <i>dnaKJ</i> -Cm ^r	This study
RAM3321	RAM3027 Δ <i>lon</i> ::Km ^r	This study
RAM3322	RAM3028 Δ <i>lon</i> ::Km ^r	This study
RAM3347	RAM3027 Δ <i>clpP</i> ::Km ^r	This study
RAM3348	RAM3028 Δ <i>clpP</i> ::Km ^r	This study
RAM3337	RAM3027 Δ <i>queF</i> ::Km ^r	This study
RAM3338	RAM3027 <i>relA</i> ⁺ Δ <i>queF</i> ::Km ^r	This study
RAM3341	RAM3028 Δ <i>queF</i> ::Km ^r	This study
RAM3342	RAM3028 <i>relA</i> ⁺ Δ <i>queF</i> ::Km ^r	This study
RAM3486	RAM3027 <i>relA</i> ^{QUAD} Δ <i>queF</i> ::Km ^r	This study
RAM3488	RAM3028 <i>relA</i> ^{QUAD} Δ <i>queF</i> ::Km ^r	This study
RAM3673	RAM1292 <i>relA</i> ⁺ Δ <i>queF</i> ::scar Δ <i>dnaJ</i> ::Km ^r	This study
RAM3674	RAM1292 <i>relA</i> ^{QUAD} Δ <i>queF</i> ::scar Δ <i>dnaJ</i> ::Km ^r	This study
RAM3133	RLG4996 <i>rrnB</i> P1::lacZ	Paul et al. (2004)
RAM3134	RLG4998 <i>lacUV5</i> ::lacZ	Paul et al. (2004)
RAM3233	RAM3133 <i>btuB</i> ::Tn10	This study
RAM3234	RAM3133 <i>rpoB58</i> <i>btuB</i> ::Tn10	This study

(Continues)

TABLE 1 | (Continued)

Strain name	Key genotype	Source
RAM3235	RAM3134 <i>btuB</i> ::Tn10	This study
RAM3236	RAM3134 <i>btuB</i> ::Tn10	This study
RAM3237	RAM3233 Δ <i>dkA</i> ::Km ^r	This study
RAM3238	RAM3234 Δ <i>dkA</i> ::Km ^r	This study
RAM3241	RAM3233 P(IPTG) <i>lac</i> :: <i>dnaKJ</i> -Cm ^r	This study
RAM3242	RAM3234 P(IPTG) <i>lac</i> :: <i>dnaKJ</i> -Cm ^r	This study
RAM3589	RAM1292 <i>relA</i> ⁺ Δ <i>queF</i> ::Km ^r	This study
RAM3591	RAM3589 Δ <i>queF</i> ::scar	This study
RAM3595	RAM3591 <i>btuB</i> ::Tn10 <i>rpoBWT</i>	This study
RAM3599	RAM3591 <i>btuB</i> ::Tn10 <i>rpoB58</i>	This study
RAM3641	RAM3595 Δ <i>dnaJ</i> ::Km ^r	This study
RAM3645	RAM3599 Δ <i>dnaJ</i> ::Km ^r	This study
RAM3657	RAM3595 Δ <i>relA</i> ::Km ^r	This study
RAM3661	RAM3599 Δ <i>relA</i> ::Km ^r	This study
RAM3675	RAM3657 Δ <i>relA</i> ::scar	This study
RAM3679	RAM3661 Δ <i>relA</i> ::scar	This study
RAM3683	RAM3675 <i>spoT</i> ::Cm ^r	This study
RAM3687	RAM3679 <i>spoT</i> ::Cm ^r	This study
RAM3691	RAM3683 Δ <i>dnaJ</i> ::Km ^r	This study
RAM3695	RAM3687 Δ <i>dnaJ</i> ::Km ^r	This study
RAM3732	RAM3027 <i>relA1</i> Δ <i>queF</i> ::Km ^r	This study
RAM3733	RAM3027 <i>relA</i> ⁺ Δ <i>queF</i> ::Km ^r	This study
RAM3734	RAM3027 <i>relA</i> ^{QUAD} Δ <i>queF</i> ::Km ^r	This study
RAM3735	RAM3028 <i>relA1</i> Δ <i>queF</i> ::Km ^r	This study
RAM3736	RAM3028 <i>relA</i> ⁺ Δ <i>queF</i> ::Km ^r	This study
RAM3737	RAM3028 <i>relA</i> ^{QUAD} Δ <i>queF</i> ::Km ^r	This study

of *relA1* was done by analyzing the *relA* region by PCR; DNA amplified from the *relA1* strain migrates slower due to the presence of a roughly 1.3-kb long IS2 element. Lambda phage was used to determine whether the *rpoB58* mutation restores phage replication, thus making the strain sensitive to lambda in the absence of DnaJ.

β -galactosidase assays: Expression of the chromosomally-localized *rrnB* P1::lacZ and *lacUV5*::lacZ gene fusion constructs (Paul et al. 2004) was measured by assaying for the production of β -galactosidase from LB or M63-glucose cultures grown overnight at 37°C. These assays were carried out from three independent cultures in duplicate by the method of Miller (1992). *p*-values of beta-galactosidase activity differences between strains were determined by an unpaired two-tailed Student's *t*-test.

Protein synthesis inhibition: Strains to be tested were streaked onto LBA plates and incubated overnight at 30°C to permit growth of temperature-sensitive Δ *dnaJ* strains. To slow protein synthesis, colonies were then streaked to LBA plates containing either 0.625, 1.25, or 2.5 μ g mL⁻¹ chloramphenicol and incubated for 36 h at the same permissive temperature of 30°C (Figure S7). Growth on LBA plates with 2.5 μ g mL⁻¹ chloramphenicol was significantly inhibited and thus not used in the subsequent experiment. To test whether temperature sensitivity was affected by the reduction of protein synthesis, colonies were then streaked from each chloramphenicol plate onto LBA plates containing either 0.625, 1.25 μ g mL⁻¹, or no chloramphenicol and incubated for 24 h at a non-permissive temperature of 38.5°C.

Microscopy techniques: Stationary phase bacterial cultures were grown overnight. Exponential phase cultures were obtained by diluting stationary phase cultures 1:100 in 3 mL broth and growing at 37°C to mid-log (OD₆₀₀ of approximately 0.4). Aliquots of each culture were put on ice until imaging. 10 μ L of iced culture was dropped onto a clean glass slide and topped with a cover slip. Slides were imaged immediately by phase contrast using a Zeiss Axioskop light microscope (Leica Microsystems, Germany) equipped with a 100 $\times$ 1.30NA Oil Plan NEOFLUAR objective lens, phase contrast optics, and a Hamamatsu ORCA-Flash4.0 LT3 Digital CMOS camera system. Images were acquired with the accompanying HCImage Live software. Custom Python scripts (modified from <https://github.com/strocha1/CellSize.git>) were used to filter the images and measure cell width and length in pixels. 1 μ m is equal to 15.5 pixels in the image. Image artefacts not filtered by the script were manually removed from the dataset, as was data for cells with visible septa formation. Custom R scripts were used to aggregate the data and calculate average cell length and width in each sample. Averages were calculated from two replicates for each sample, with > 30 cells per replicate. Each average represents > 60 cells across both replicates of the sample.

Scanning electron microscopic (SEM) analysis was conducted with stationary phase cultures. Cells were pelleted and washed using 20 mM phosphate buffer solution (pH 7.0). After pelleting and washing twice, cells were fixed in 2% glutaraldehyde before imaging by SEM using an in-house microscopy facility. Cell dimensions (length and width) were analyzed using ImageJ software.

Author Contributions

Melody Yeh: methodology, investigation, validation, formal analysis, data curation. **Keilen Kelly:** investigation, methodology, software,

formal analysis, writing – review and editing. **Rajeev Misra:** conceptualization, investigation, funding acquisition, writing – original draft, writing – review and editing, methodology, formal analysis, project administration, resources, supervision, validation.

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Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

References

- Ades, S. E., L. E. Connolly, B. M. Alba, and C. A. Gross. 1999. "The *Escherichia coli* σ^E -Dependent Extracytoplasmic Stress Response Is Controlled by the Regulated Proteolysis of an Anti- σ Factor." *Genes & Development* 13: 2449–2461. <https://doi.org/10.1101/gad.13.18.2449>.
- Balchin, D., M. Hayer-Hartl, and F. U. Hartl. 2020. "Recent Advances in Understanding Catalysis of Protein Folding by Molecular Chaperones." *FEBS Letters* 594: 2770–2781. <https://doi.org/10.1002/1873-3468.13844>.
- Barker, M. M., T. Gall, and R. L. Gourse. 2001. "Mechanism of Regulation of Transcription Initiation by ppGpp. II. Models for Positive Control Based on Properties of RNAP Mutants and Competition for RNAP." *Journal of Molecular Biology* 305: 689–702. <https://doi.org/10.1006/jmbi.2000.4328>.
- Blaby, I. K., B. J. Lyons, E. Wroclawska-Hughes, et al. 2012. "Experimental Evolution of a Facultative Thermophile From a Mesophilic Ancestor." *Applied and Environmental Microbiology* 78, no. 1: 144–155. <https://doi.org/10.1128/AEM.05773-11>.
- Bruel, N., M.-P. Castani -Cornet, A.-M. Cirinesi, et al. 2012. "Hsp33 Controls Elongation Factor-Tu Stability and Allows *Escherichia coli* Growth in the Absence of Major DnaK and Trigger Factor Chaperones." *Journal of Biological Chemistry* 287: 44435–44446. <https://doi.org/10.1074/jbc.M112.418525>.
- Bukau, B., and G. C. Walker. 1989. "Cellular Defects Caused by the Deletion of *Escherichia coli* Gene *dnaK* Indicates Roles for Heat Shock Proteins in Normal Metabolism." *Journal of Bacteriology* 171: 2337–2346. <https://doi.org/10.1128/jb.171.5.2337-2346.1989>.
- Calloni, G., T. Chen, S. M. Schermann, et al. 2012. "DnaK Functions as a Central Hub in the *E. coli* Chaperone Network." *Cell Reports* 1: 251–264. <https://doi.org/10.1016/j.celrep.2011.12.007>.
- Casadaban, M. J. 1976. "Transposition and Fusion of the *Lac* Gene to Selected Promoters in *Escherichia coli* Using Bacteriophage Lambda and Mu." *Journal of Molecular Biology* 104: 541–555. [https://doi.org/10.1016/0022-2836\(76\)90119-4](https://doi.org/10.1016/0022-2836(76)90119-4).
- Cashel, M., and J. Gallant. 1969. "Two Compounds Implicated in the Function of the *RC* Gene of *Escherichia coli*." *Nature* 221: 838–841. <https://doi.org/10.1038/221838a0>.
- Cho, H.-J., and R. Misra. 2021. "Mutational Activation of Antibiotic Resistant Pathways in the Absence of Two Major Antibiotic Efflux Systems of *Escherichia coli*." *Journal of Bacteriology* 203: e0010921. <https://doi.org/10.1128/jb.0010921>.
- Conrad, T. M., M. Frazier, A. R. Joyce, et al. 2010. "RNA Polymerase Mutants Found Through Adaptive Evolution Reprogram *Escherichia*

- coli* for Optimal Growth on Minimal Media." *Proceedings of the National Academy of Sciences of the United States of America* 107: 20500–20505. <https://doi.org/10.1073/pnas.0911253107>.
- Danese, P. N., and T. J. Silhavy. 1997. "The σ^E and the Cpx Signal Transduction Systems Control the Synthesis of Periplasmic Protein-Folding Enzymes in *Escherichia coli*." *Genes & Development* 11: 1183–1193. <https://doi.org/10.1101/gad.11.9.1183>.
- Dragosits, M., V. Mozhayskiy, S. Quinones-Soto, J. Park, and I. Tagkopoulos. 2013. "Evolutionary Potential, Cross-Stress Behavior and the Genetic Basis of Acquired Stress Resistance in *Escherichia coli*." *Molecular Systems Biology* 9: 643. <https://doi.org/10.1038/msb.2012.76>.
- Fujiwara, K., Y. Ishihama, K. Nakahigashi, T. Soga, and H. Taguchi. 2010. "A Systematic Survey on In Vivo Obligate Chaperonin-Dependent Substrates." *EMBO Journal* 29: 1552–1564. <https://doi.org/10.1038/emboj.2010.52>.
- Fujiwara, K., and H. Taguchi. 2007. "Filamentous Morphology in GroE-Depleted *Escherichia coli* Induced by Impaired Folding of FtsE." *Journal of Bacteriology* 189: 5860–5866. <https://doi.org/10.1128/jb.00493-07>.
- Gerken, H., O. P. Leiser, D. Bennion, and R. Misra. 2010. "Involvement and Necessity of the Cpx Regulon in the Event of Aberrant β -barrel Outer Membrane Protein Assembly." *Molecular Microbiology* 75: 1033–1046. <https://doi.org/10.1111/j.1365-2958.2009.07042.x>.
- Grossman, A. D., J. W. Erickson, and C. A. Gross. 1984. "The *htpR* Gene Product of *Escherichia coli* Is a Sigma Factor for Heat-Shock Promoters." *Cell* 38: 383–390. [https://doi.org/10.1016/0092-8674\(84\)90493-8](https://doi.org/10.1016/0092-8674(84)90493-8).
- Hoffman, H., and M. E. Frank. 1963. "Temperature Limits, Genealogical Origin, Developmental Course, and Ultimate of Heat-Induced Filaments in *Escherichia coli* Microcultures." *Journal of Bacteriology* 85: 1221–1234. <https://doi.org/10.1128/jb.85.6.1221-1234.1963>.
- Imholz, N. C. E., M. J. Noga, N. J. F. van den Broek, and G. Bokinsky. 2020. "Calibrating the Bacterial Growth Rate Speedometer: A Re-Evaluation of the Relationship Between Basal ppGpp, Growth, and RNA Synthesis in *Escherichia coli*." *Frontiers in Microbiology* 11: 574872. <https://doi.org/10.3389/fmicb.2020.574872>.
- Johansen, J., A. A. Rasmussen, M. Overgaard, and P. Valentin-Hansen. 2006. "Conserved Small Non-Coding RNAs That Belong to the σ^E Regulon: Role in Down-Regulation of Outer Membrane Proteins." *Journal of Molecular Biology* 364: 1–8. <https://doi.org/10.1016/j.jmb.2006.09.004>.
- Kang, P. J., and E. A. Craig. 1990. "Identification and Characterization of New *Escherichia coli* Gene That Is Dosage Dependent Suppressor of a *dnaK* Deletion Mutation." *Journal of Bacteriology* 172: 2055–2064. <https://doi.org/10.1128/jb.172.4.2055-2064.1990>.
- Kaspy, I., E. Rotem, N. Weiss, I. Ronin, N. Q. Balaban, and G. Glaser. 2015. "HipA-Mediated Antibiotic Persistence via Phosphorylation of the Glutamyl-tRNA-Synthetase." *Nature Communications* 4: 3001. <https://doi.org/10.1038/ncomms4001>.
- Kerner, M. J., D. J. Naylor, Y. Ishihama, et al. 2005. "Proteome-Wide Analysis of Chaperonin-Dependent Protein Folding in *Escherichia coli*." *Cell* 122: 209–220. <https://doi.org/10.1016/j.cell.2005.05.028>.
- Kishimoto, T., L. Iijima, M. Tatsumi, et al. 2010. "Transition From Positive to Neutral Mutation in Fixation Along Continuing Rising Fitness in Thermal Adaptive Evolution." *PLoS Genetics* 10: e1001164. <https://doi.org/10.1371/journal.pgen.1001164>.
- Kosaka, T., Y. Nakajima, A. Ishii, et al. 2019. "Capacity for Survival in Global Warming: Adaptation of Mesophiles to the Temperature Upper Limit." *PLoS One* 14: e0215614. <https://doi.org/10.1371/journal.pone.0215614>.
- Lemke, J. J., P. Sanchez-Vazquez, H. L. Burgos, and R. L. Gourse. 2011. "Direct Regulation of *Escherichia coli* Ribosomal Protein Promoters by the Transcription Factors ppGpp and DksA." *Proceedings of the National Academy of Sciences of the United States of America* 108: 5712–5717. <https://doi.org/10.1073/pnas.1019383108>.
- Li, G.-W., D. Burkhardt, C. Gross, and J. S. Weissman. 2014. "Quantifying Absolute Protein Synthesis Rates Reveals Principles Underlying Allocation of Cellular Resources." *Cell* 157: 624–635. <https://doi.org/10.1016/j.cell.2014.02.033>.
- Liberek, K., C. Georgopoulos, and M. Zylick. 1988. "Role of *Escherichia coli* DnaK and DnaJ Heat Shock Proteins in the Initiation of Bacteriophage λ DNA Replication." *Proceedings of the National Academy of Sciences of the United States of America* 85: 6632–6636. <https://doi.org/10.1073/pnas.85.18.6632>.
- Luan, G., G. Bao, Z. Lin, et al. 2015. "Comparative Genome Analysis of a Thermotolerant *Escherichia coli* Obtained by Genome Replication Engineering Assisted Continuous Evolution (GREACE) and Its Parent Strain Provides New Understanding of Microbial Heat Tolerance." *New Biotechnology* 32: 732–738. <https://doi.org/10.1016/j.nbt.2015.01.013>.
- Maurizi, M. R., W. P. Clark, Y. Katayama, et al. 1990. "Sequence and Structure of ClpP, the Proteolytic Component of the ATP-Dependent Clp Protease of *Escherichia coli*." *Journal of Biological Chemistry* 265: 12536–12545. [https://doi.org/10.1016/S0021-9258\(19\)38378-4](https://doi.org/10.1016/S0021-9258(19)38378-4).
- Maurizi, M. R., P. Trisler, and S. Gottesman. 1985. "Insertional Mutagenesis of *Lon* Gene of *Escherichia coli*: *Lon* Is Dispensable." *Journal of Bacteriology* 164: 1124–1135. <https://doi.org/10.1128/jb.164.3.1124-1135.1985>.
- Mayer, M. P. 2021. "The Hsp70-Chaperone Machinery in Bacteria." *Frontiers in Molecular Biosciences* 8: 694012. <https://doi.org/10.3389/fmolb.2021.694012>.
- McGuire, B. E., and F. E. Nano. 2023. "Whole-Genome Sequence Analysis Two Heat-Evolved *Escherichia coli* Strains." *BMC Genomics* 24: 154. <https://doi.org/10.1186/s12864-023-09266-9>.
- Meriin, A. B., M. Mense, J. D. Colbert, et al. 2012. "A Novel Approach to Recovery of Function of Mutant Proteins by Slowing Down Translation." *Journal of Biological Chemistry* 287: 34264–34272. <https://doi.org/10.1074/jbc.M112.397307>.
- Metzger, S., G. Schreiber, E. Aizenman, M. Cashel, and G. Glaser. 1989. "Characterization of the *relA1* Mutation and a Comparison of *relA1* With New *relA* Null Alleles in *Escherichia coli*." *Journal of Biological Chemistry* 264: 21146–21152. [https://doi.org/10.1016/S0021-9258\(19\)30059-6](https://doi.org/10.1016/S0021-9258(19)30059-6).
- Miller, J. H. 1992. *A Short Course in Bacterial Genetics: A Laboratory Manual and Handbook for Escherichia coli and Related Bacteria*, 71–74. Cold Spring Harbor Laboratory Press. <https://doi.org/10.1002/jobm.3620330412>.
- Milon, P., E. Tishenko, J. Tomšić, et al. 2006. "The Nucleotide Binding Site of Bacterial Translation Initiation Factor 2 (IF2) as a Metabolic Sensor." *Proceedings of the National Academy of Sciences of the United States of America* 103: 13962–13967. <https://doi.org/10.1073/pnas.0606384103>.
- Mogk, A., T. Tomoyasu, P. Goloubinoff, et al. 1992. "Identification of Thermolabile *Escherichia coli* Proteins: Prevention and Reversion of Aggregation by DnaK and ClpB." *EMBO Journal* 18: 6934–6949. <https://doi.org/10.1093/emboj/18.24.6934>.
- Murata, M., H. Fujimoto, K. Nishimura, et al. 2011. "Molecular Strategy for Survival at Critical High Temperature in *Escherichia coli*." *PLoS One* 6: e20063. <https://doi.org/10.1371/journal.pone.0020063>.
- Murphy, H., and M. Cashel. 2003. "Isolation of RNA Polymerase Suppressors of a (p)ppGpp Deficiency." *Methods in Enzymology* 371: 596–601. [https://doi.org/10.1016/s0076-6879\(03\)71044-1](https://doi.org/10.1016/s0076-6879(03)71044-1).
- Paek, K.-H., and G. C. Walker. 1987. "*Escherichia coli* *dnaK* Null Mutants Are Inviably at High Temperature." *Journal of Bacteriology* 169: 283–290. <https://doi.org/10.1128/jb.169.1.283-290.1987>.

- Paul, B. J., M. M. Barker, W. Ross, et al. 2004. "DksA: A Critical Component of the Transcription Initiation Machinery That Potentiates the Regulation of rRNA Promoters by ppGpp and the Initiating NTP." *Cell* 118: 311–322. <https://doi.org/10.1016/j.cell.2004.07.009>.
- René, O., and J.-H. Alix. 2010. "Late Steps of Ribosome Assembly in *Escherichia coli* Are Sensitive to a Severe Heat Stress but Are Assisted by the HSP70 Chaperone Machinery." *Nucleic Acids Research* 39: 1855–1867. <https://doi.org/10.1093/nar/gkq1049>.
- Rodríguez-Verdugo, A., D. Carrillo-Cisneros, A. González-González, B. S. Gaut, and A. F. Bennett. 2014. "Different Tradeoff Results From Alternate Genetic Adaptation to a Common Environment." *Proceedings of National Academy of Sciences, USA* 111: 12121–12126. <https://doi.org/10.1073/pnas.1406886111>.
- Ron, E. Z., and B. D. Davis. 1971. "Growth Rate of *Escherichia coli* at Elevated Temperatures: Limitation by Methionine." *Journal of Bacteriology* 107: 391–396. <https://doi.org/10.1128/jb.107.2.391-396.1971>.
- Ross, W., P. Sanchez-Vazquez, A. Y. Chen, J.-H. Lee, H. L. Burgos, and R. L. Gourse. 2016. "ppGpp Binding to a Site at the RNAP-DksA Interface Accounts for Its Dramatic Effects on Transcription Initiation During the Stringent Response." *Molecular Cell* 62: 811–823. <https://doi.org/10.1016/j.molcel.2016.04.029>.
- Rudolph, B., K. M. Gebendorfer, J. Buchner, and J. Winter. 2010. "Evolution of *Escherichia coli* at High Temperatures." *Journal of Biological Chemistry* 285: 19029–19034. <https://doi.org/10.1074/jbc.M110.103374>.
- Sanchez-Vazquez, P., C. N. Dewey, N. Kitten, W. Ross, and R. L. Gourse. 2019. "Genome-Wide Effects on *Escherichia coli* Transcription From ppGpp Binding to Its Two Sites on RNA Polymerase." *Proceedings of the National Academy of Sciences of the United States of America* 116: 8310–8319. <https://doi.org/10.1073/pnas.1819682116>.
- Schäfer, H., B. Beckert, C. K. Frese, et al. 2020. "The Alarmones (p) ppGpp Are Part of the Heat Shock Response of *Bacillus subtilis*." *PLoS Genetics* 16: e1008275. <https://doi.org/10.1371/journal.pgen.1008275>.
- Schramm, F. D., K. Schroeder, and K. Jonas. 2020. "Protein Aggregation in Bacteria." *FEMS Microbiology Reviews* 44: 54–72. <https://doi.org/10.1093/femsre/fuz026>.
- Sell, S. M., C. Eisen, D. Ang, M. Zylicz, and C. Georgopoulos. 1990. "Isolation and Characterization of Null Mutants of *Escherichia coli*." *Journal of Bacteriology* 172: 4827–4835. <https://doi.org/10.1128/jb.172.9.4827-4835.1990>.
- Shiver, A. L., H. Osadnik, J. M. Peters, et al. 2021. "Chemical-Genetic Interrogation of RNA Polymerase Mutants Reveals Structure-Function Relationships and Physiological Tradeoffs." *Molecular Cell* 81: 2201–2215. <https://doi.org/10.1016/j.molcel.2021.04.027>.
- Silhavy, T. J., M. L. Berman, and L. W. Enquist. 1984. *Experiments With Gene Fusions*. Cold Spring Harbor Laboratory Press.
- Siller, E., D. C. DeZwaan, J. F. Anderson, B. C. Freeman, and J. M. Barral. 2010. "Slowing Protein Translation Speed Enhances Eukaryotic Protein Folding Efficiency." *Journal of Molecular Biology* 396: 1310–1318. <https://doi.org/10.1016/j.jmb.2009.12.042>.
- Sinha, A. K., and K. S. Winther. 2021. "The RelA Hydrolase Domain Acts as Molecular Switch for (p)ppGpp Synthesis." *Communications Biology* 4: 434. <https://doi.org/10.1038/s42003-021-01963-z>.
- Stent, G. S., and S. Brenner. 1961. "A Genetic Locus for the Regulation of Ribonucleic Acid Synthesis." *Proceedings of the National Academy of Sciences of the United States of America* 47: 2005–2014. <https://doi.org/10.1073/pnas.47.12.2005>.
- Strauch, K. L., K. Johnson, and J. Beckwith. 1989. "Characterization of *degP*, a Gene Required for Proteolysis in the Cell Envelope and Essential for Growth of *Escherichia coli* at High Temperature." *Journal of Bacteriology* 171: 2689–2696. <https://doi.org/10.1128/jb.171.5.2689-2696.1989>.
- Straus, D., W. Walter, and C. A. Gross. 1990. "DnaK, DnaJ, and GrpE Heat Shock Proteins Negatively Regulate Heat Shock Gene Expression by Controlling the Synthesis and Stability of σ^{32} ." *Genes & Development* 4: 2202–2209. <https://doi.org/10.1101/gad.4.12a.2202>.
- Sunshine, M., M. Feiss, J. Stuart, and J. Yochem. 1977. "A New Host Gene (*groPC*) Necessary for Lambda DNA Replication." *Molecular and General Genetics* 151: 27–34. <https://doi.org/10.1007/BF00446909>.
- Szalewska-Palasz, A., L. U. M. Johansson, L. M. D. Bernardo, et al. 2007. "Properties of RNA Polymerase Bypass Mutants. Implications for the Role of ppGpp and Its Co-Factor DksA in Controlling Transcription Dependent on σ^{54} ." *Journal of Biological Chemistry* 282: 18046–18056. <https://doi.org/10.1074/jbc.M610181200>.
- Tao, H., C. Bausch, C. Richmond, F. R. Blattner, and T. Conway. 1999. "Functional Genomics: Expression Analysis of *Escherichia coli* Growing on Minimal and Rich Media." *Journal of Bacteriology* 181: 6425–6440. <https://doi.org/10.1128/jb.181.20.6425-6440.1999>.
- Tenaillon, O., A. Rodríguez-Verdugo, R. Gaut, et al. 2012. "The Molecular Diversity of Adaptive Convergences." *Science* 335: 457–461. <https://doi.org/10.1126/science.1212986>.
- Tomoyasu, T., A. Mogk, H. Langen, P. Goloubinoff, and B. Bukau. 2001. "Genetic Dissection of the Roles of Chaperones and Proteases in Protein Folding and Aggregation in the *Escherichia coli* Cytosol." *Molecular Microbiology* 40: 397–413. <https://doi.org/10.1046/j.1365-2958.2001.02383.x>.
- Tomoyasu, T., T. Ogura, T. Tatsuta, and B. Bukau. 1998. "Levels of DnaK and DnaJ Provide Tight Control of Heat Shock Genes Expression and Protein Repair in *Escherichia coli*." *Molecular Microbiology* 30: 567–581. <https://doi.org/10.1046/j.1365-2958.1998.01090.x>.
- Trautinger, B. W., and R. G. Lloyd. 2002. "Modulation of DNA Repair by Mutations Flanking the DNA Channel Through RNA Polymerase." *EMBO Journal* 21: 6944–6953. <https://doi.org/10.1093/emboj/cdf654>.
- Ueguchi, C., T. Shiozawa, M. Kakeda, H. Yamada, and T. Mizuno. 1995. "A Study of the Double Mutation of *dnaJ* and *cbpA*, Whose Gene Products Function as Molecular Chaperones in *Escherichia coli*." *Journal of Bacteriology* 177: 3894–3896. <https://doi.org/10.1128/jb.177.13.3894-3896.1995>.
- Utrilla, J., E. J. O'Brian, K. Chen, et al. 2016. "Global Rebalancing of Cellular Resources by Pleiotropic Point Mutations Illustrates a Multi-Scale Mechanism of Adaptive Evolution." *Cell Systems* 2: 260–271. <https://doi.org/10.1016/j.cels.2016.04.003>.
- Uzan, M., and A. Danchin. 1976. "A Rapid Test for the *relA* Mutation in *E. coli* (1976)." *Biochemical and Biophysical Research Communications* 69: 751–758. [https://doi.org/10.1016/0006-291x\(76\)90939-6](https://doi.org/10.1016/0006-291x(76)90939-6).
- van de Vossenberg, J. L. C. M., T. Ubbink-Kok, M. G. L. Elferink, A. J. M. Driessen, and W. N. Konings. 1995. "Ion Permeability of the Cytoplasmic Membrane Limits the Maximum Growth Temperature of Bacteria and Archaea." *Molecular Microbiology* 18: 925–932. <https://doi.org/10.1111/j.1365-2958.1995.18050925.x>.
- van Derlinden, E., K. Bernaerts, and J. F. Van Impe. 2008. "Dynamics of *Escherichia coli* at Elevated Temperatures: Effect of Temperature History and Medium." *Journal of Applied Microbiology* 104: 438–453. <https://doi.org/10.1111/j.1365-2672.2007.03592.x>.
- Werner, J., and R. Misra. 2005. "YaeT Affects the Assembly of Lipid-Dependent and Lipid-Independent Outer Membrane Proteins of *Escherichia coli*." *Molecular Microbiology* 57: 1450–1459. <https://doi.org/10.1111/j.1365-2958.2005.04775.x>.
- Xiao, H., M. Kalman, K. Ikehara, S. Zemel, G. Glaser, and M. Cashel. 1991. "Residual Guanine 3', 5'-Bispyrophosphate Synthetic Activity of *relA* Null Mutants Can Be Eliminated by *Spot* Null Mutations." *Journal*

of *Biological Chemistry* 266: 5980–5990. [https://doi.org/10.1016/S0021-9258\(19\)67694-5](https://doi.org/10.1016/S0021-9258(19)67694-5).

Yang, X., and E. E. Ishiguro. 2003. “Temperature Sensitive Growth and Decreased Thermotolerance Associated With *relA* Mutations in *Escherichia coli*.” *Journal of Bacteriology* 185: 5765–5771. <https://doi.org/10.1128/jb.185.19.5765-5771.2003>.

Zhang, Y. M., and C. O. Rock. 2008. “Membrane Lipid Homeostasis in Bacteria.” *Nature Reviews Microbiology* 6: 222–232. <https://doi.org/10.1038/nrmicro1839>.

Zhou, Y. N., and J. D. Jin. 1998. “The *rpoB* Mutants Destabilizing Initiation Complexes at Stringently Controlled Promoters Behave Like “Stringent” RNA Polymerase in *Escherichia coli*.” *Proceeding of National Academy of Sciences, USA* 95: 2908–2913. <https://doi.org/10.1073/pnas.95.6.2908>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.